



College of Sciences
Department of Computer Science

Course Syllabus

Course Title
**Introduction to Database Management
Systems**

Course Code
1501263

Fall Semester- 2023/2024

Course Syllabus

Course Title	Introduction to Database Management Systems		
Course Code	1501263		
Credit Hours	3		
Pre-requisite(s)	1501215 Data Structures		
Co-requisite(s)	None		
Semester	Fall	Year	2023-2024
Instructors Name	Dr. Osman Abul		
Office Location	M5-209A		
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Lecture Times			
Office Hours	TBA on Blackboard		

Course Description (as in the catalogue):

This course explores how databases are designed, implemented and used. The course emphasizes the basic concepts/terminology of the relational model and applications. The students will learn database design concepts, data models (Entity-Relationship and Relational Model), SQL, functional dependencies and normal forms. The students will gain experience working with a database management system.

Course Objectives/Goals:

The goals of this course are to:

1. Understand the components and the capabilities of a database system
2. Provide an understanding of, and hands on experience in, designing and developing a database system.
3. Provide an understanding of the relational database model.
4. Provide familiarity with a commercial database management system.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

- A. Identify the components and the capabilities of a database management system.
- B. Discuss the basic principles of the relational model, enforcing constraints, handling violations and querying using relational algebra.
- C. Create and manipulate databases and retrieve data from a relational database model using SQL queries.
- D. Analyze and design a relational database system using ER model that meets users' data requirements.
- E. Evaluate and refine a database design through functional dependencies and normal forms.
- F. Work as part of a team to design and develop a database system using a DBMS.

Alignment of Course Student Learning Outcomes to the BSc in CS Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	Analyze	- Analyze and design a relational database system using ER model that meets users' data requirements.
	Apply	- Create and manipulate databases and retrieve data from a relational database model using SQL queries.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Design	- Analyze and design a relational database system using ER model that meets users' data requirements.
	Implement	- Create and manipulate databases and retrieve data from a relational database model using SQL queries.
	Evaluate	- Evaluate and refine a database design through functional dependencies and normal forms.
Communicate effectively in a variety of professional contexts.		- Work as part of a team to design and develop a database system using a commercial DBMS
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		- Work as part of a team to design and develop a database system using a commercial DBMS.
Apply computer science theory and software development fundamentals to produce computing-based solutions.	Theory	- Discuss the basic principles of the relational model, enforcing constraints, handling violations and querying using relational algebra. - Identify the components and the capabilities of a database management system.
	Development	- Create and manipulate databases and retrieve data from a relational database model using SQL queries. - Work as part of a team to design and develop a database system using a commercial DBMS.

Alignment of Course Student Learning Outcomes to the BSc in IT Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	Analyze	- Analyze and design a relational database system using ER model that meets users' data requirements.
	Apply	- Create and manipulate databases and retrieve data from a relational database model using SQL queries.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Design	- Analyze and design a relational database system using ER model that meets users' data requirements.
	Implement	- Create and manipulate databases and retrieve data from a relational database model using SQL queries.
	Evaluate	- Evaluate and refine a database design through functional dependencies and normal forms.
Communicate effectively in a variety of professional contexts.		- Work as part of a team to design and develop a database system using a commercial DBMS
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		- Work as part of a team to design and develop a database system using a commercial DBMS.
Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based systems.	Identify / Analyze	- Identify the components and the capabilities of a database management system. - Analyze and design a relational database system using ER model that meets users' data requirements. - Discuss the basic principles of the relational model, enforcing constraints, handling violations and querying using relational algebra.
	Development	- Create and manipulate databases and retrieve data from a relational database model using SQL queries. - Work as part of a team to design and develop a database system using a commercial DBMS.

Weekly Distribution of Course Topics/Contents

Week	Topic	Exercises	HW	Course SLO
1.	Introduction: Databases and Database Users.	iTBL 1	HW/Quiz1	A

Week	Topic	Exercises	HW	Course SLO
2.	Database Concepts and Architecture: Data Models; Schemes and Instances; and DBMS Architecture	iTBL 2		B
3.	Database Concepts and Architecture: Program Data Independence; Database Languages and Interfaces			B
4.	The Relational Data Model: Relational DM Concepts; Relational Constraints and DB Schemas.	iTBL 3	HW/Quiz2	B
5.	Relational Algebra: Basic Relational Algebra Operations			B
6.	SQL: Data Definition in SQL; Queries in SQL; Select Statements in SQL	iTBL 4		C
7.	SQL: Select Statements in SQL; Nested queries; Insert Statements, Delete Statements			C
8.	SQL: Views in SQL; practical sessions.			C
9.		Midterm Exam		
10.	Data Modeling using ER Model: High-level Conceptual Data Models for Database Design; Entity types, sets, attributes and keys;	iTBL 5	HW/Quiz3	D
11.	Data Modeling using ER Model: Entity types, sets, attributes and keys; Relationships and Structural Constraints	iTBL 6		D
12.	Data Modeling using ER Model: Enhanced Entity-Relationship Model	iTBL 7		D
13.	Mapping a Conceptual Design into a Logical Design	iTBL 8	Project	E
14.	Relational Database Design: Functional Dependencies; Normal Forms	iTBL 9		E
15.	Relational Database Design: Functional Dependencies; Normal Forms			E
16.	Project presentations and Discussions			F
17.	Final Exam			

* In the comments, you can add the relevant chapter or notes, etc.

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date	Comments
HW/Quiz 1		
HW/Quiz 2		
HW/Quiz 3		

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Assignment/Quizzes		10
iTBLs		15
Midterm exam		25
Project		15
Final Exam		35
Total		100

** You can modify / add other tools relevant to the course.

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Identify the components and the capabilities of a database management system.	Lectures Exercises	Assignments/Quizzes iTBLs Midterm Exam Final Exam	Marks
Discuss the basic principles of the relational model, enforcing constraints, handling violations and querying using relational algebra	Lectures Exercises	Assignments/Quizzes iTBLs Midterm Exam Final Exam	Marks
Create and manipulate databases and retrieve data from a relational database model using SQL queries	Lectures Exercises	Assignments/Quizzes iTBLs Midterm Exam Final Exam	Marks
Analyze and design a relational database system using ER model that meets users' data requirements	Lectures Exercises	Assignments/Quizzes iTBLs Midterm Exam Final Exam	Marks
Evaluate and refine a database design through functional dependencies and normal forms	Lectures Exercises	Assignments/Quizzes iTBLs Midterm Exam Final Exam	Marks
Work as part of a team to design and develop a database system using a commercial DBMS	Social Learning (Project and Group Exercises)	iTBLs Presentation	Marks

Teaching and Learning Resources:**Text Book(s):**

Elmasri, R. and Navathe, S. (2016). Fundamentals of Database Systems. 7th Edition. Addison-Wesley.

Recommended Readings:

- An Introduction to Database Systems, C. J. Date, 8th edition, Pearson Publisher, 2003.

Other Resources:**Computer Usage**

MySQL DBMS.

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Assignments:

Students will have 3 assignments as described in the weekly schedule.

Project:

Each project team will consist of two to four members. One member will be designated as the project team leader; this person will be responsible for coordinating the activities of the team. The duties of the members of each team will be decided by the team as a whole and need not be equal on each phase of the project, but work load must/should somehow balance up equally towards the end of the semester. Each team will specify, design, and implement, a small database application.

In-class Activities/Teaching Methodology:

The course includes lectures, lab practice, assignments, iTBLs (social learning), group projects (report & demo).

in-class Team-Based Learning (iTBL):

Students are divided into teams. After each chapter/lecture, teams are given an exercise to discuss and solve collaboratively. iTBL enhances the problem solving skills, while developing the interpersonal skills.

Lab Sessions:

Some of the classes will be held in the lab to introduce the packages/systems (MySQL DBMS) that will be used during the term. The students will be asked to design and implement their projects using MySQL DBMS.